Donato **Meoli**

Research Fellow · Mathematical Optimization · Symbolic AI · Machine Learning

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Summary

I came to numerical optimization from symbolic AI, and more precisely from Knowledge Representation and Reasoning (KR&R), i.e., expert systems, logic programming, constraint satisfaction and SAT. In my view, the two fields share one premise: the difficulty of a problem depends (often to a large extent) on how it is written. Hence, the useful move is frequently to rewrite it, either as a set of smaller problems coordinated by a master or in a different formalism for which an efficient algorithm is available; for instance, a planning problem can be stated as a Constraint Satisfaction Problem and, from there, as a propositional formula, at which point the machinery of logical reasoning applies to it.

On the numerical side, this means Lagrangian (Dantzig-Wolfe) Relaxation and Decomposition, Benders Decomposition, the Bundle Methods that solve the nondifferentiable duals they give rise to, and the parallel solution of the subproblems, which are typically many and independent by construction. Reformulation comes back inside the subproblems, as the recognition of a structure for which a specialized (often combinatorial) algorithm exists; since a decomposition method solves each of them many times with data that change rather little, much of my current work is on re-optimization, i.e., on reusing what has been computed for one instance in the solution of the next one, and on enlarging the class of subproblems that dynamic programming can solve efficiently.

I see machine learning, which connects the two sides, mainly as a way of learning heuristics, to be placed at the points where an exact algorithm makes a choice, and hence of making that algorithm faster: on the symbolic side, the heuristic that guides a search algorithm such as A* or the branching of a SAT solver; on the numerical side, the branching of a branch-and-bound whose bounds come from a specialized relaxation (e.g., a Lagrangian one). The learned part makes the choice, while the guarantees still come from the exact method. Conversely, learning problems can be written as structured optimization models, to which the same decomposition and re-optimization machinery applies.

Publications

D. Meoli, E. Wdowiak. *Climbing the Ladder: Automating Data Pipelines and Reproducing Baselines in Low-Resource Settings*. arXiv preprint, 2026. arXiv:2610.30374

A. Frangioni, C. Gentile, **D. Meoli**. *Dynamic Programming for the Single-Unit Commitment Problem with Spinning Reserves and Reactive Power*. Submitted, 2026.

D. Fioriti, A. Pampado, C. Scarpelli, G. Pasini, G. Lutzemberger, S. Barsali, D. Poli, **D. Meoli**, L. Mencarelli, A. Frangioni. *Enhancing Energy System Modelling with Advanced Mathematical Decomposition Techniques: Feasibility of Coupling SMS++ and PyPSA*. Proceedings of the 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe, pages 1–6, 2025. doi:10.1109/EEEIC/ICPSEurope64998.2025.11169291

Research Software

SMS++ • co-lead maintainer, with A. Frangioni, since 2021.

A framework for block-structured optimization, in which a model is a tree of nested blocks and an algorithm attaches to any of them; hence, a single instance can be solved monolithically, dualized, or split by scenario without rewriting the model.

aima-python • contributor since 2019.

Reference implementation of the algorithms in Russell & Norvig's *Artificial Intelligence: A Modern Approach*.

OptiML • author, since 2020.

Optimization algorithms, from line search and Newton-type methods to proximal bundle and interior point, together with the learning models trained by them, which allows one to study a method against the model it serves.

Elsewhere. *lamalunga*, a typographic theme for slides and printed pages in which the proportions, from the margins to the sizes of the type, are powers of the golden ratio, and the palette is computed from a single color by rotating its hue, with the contrast of the text checked against accessibility thresholds; *qpsolvers*, a single Python interface to the Quadratic Programming solvers, to which I contributed the lower and upper bounds on the variables and of whose citation I am a co-author; and earlier work in symbolic reasoning and knowledge representation: *NeuroSAT*, neuro-symbolic branching heuristics for satisfiability; *PESAD* and *BeerEX*, expert systems in logic programming and in rules with fuzzy reasoning; *WS4J* and *WNAffect*, semantic relatedness and affect over a lexical ontology.

Experience

Università di Pisa — *Dipartimento di Informatica, DESTeC*

Research Fellow — *Dec. 2021 - Present*

- Decomposition methods and solvers for large-scale optimization, and core development of SMS++, within the AUTENS, RESILIENT and PowerMax research projects.
- Applied to planning and scheduling under uncertainty, i.e., to models on which a general-purpose MILP solver typically does not close the instance in reasonable time.

Exprivia S.p.A. — *Italy*

Data Scientist — *Sept. 2018 - Feb. 2019*

- Learning models for predictive maintenance on multivariate time series, for image classification, and for real-time detection and counting.

Reports

Support Vector Machines — *Università di Pisa*

A Comparative Study of the Formulations from an Optimization Point of View — *June 2021*

Education

Università di Pisa — *Italy*

MSc in Computer Science, Artificial Intelligence

Università degli Studi di Bari "Aldo Moro" — *Italy*

BSc in Computer Science

Grade: 110/110 cum laude